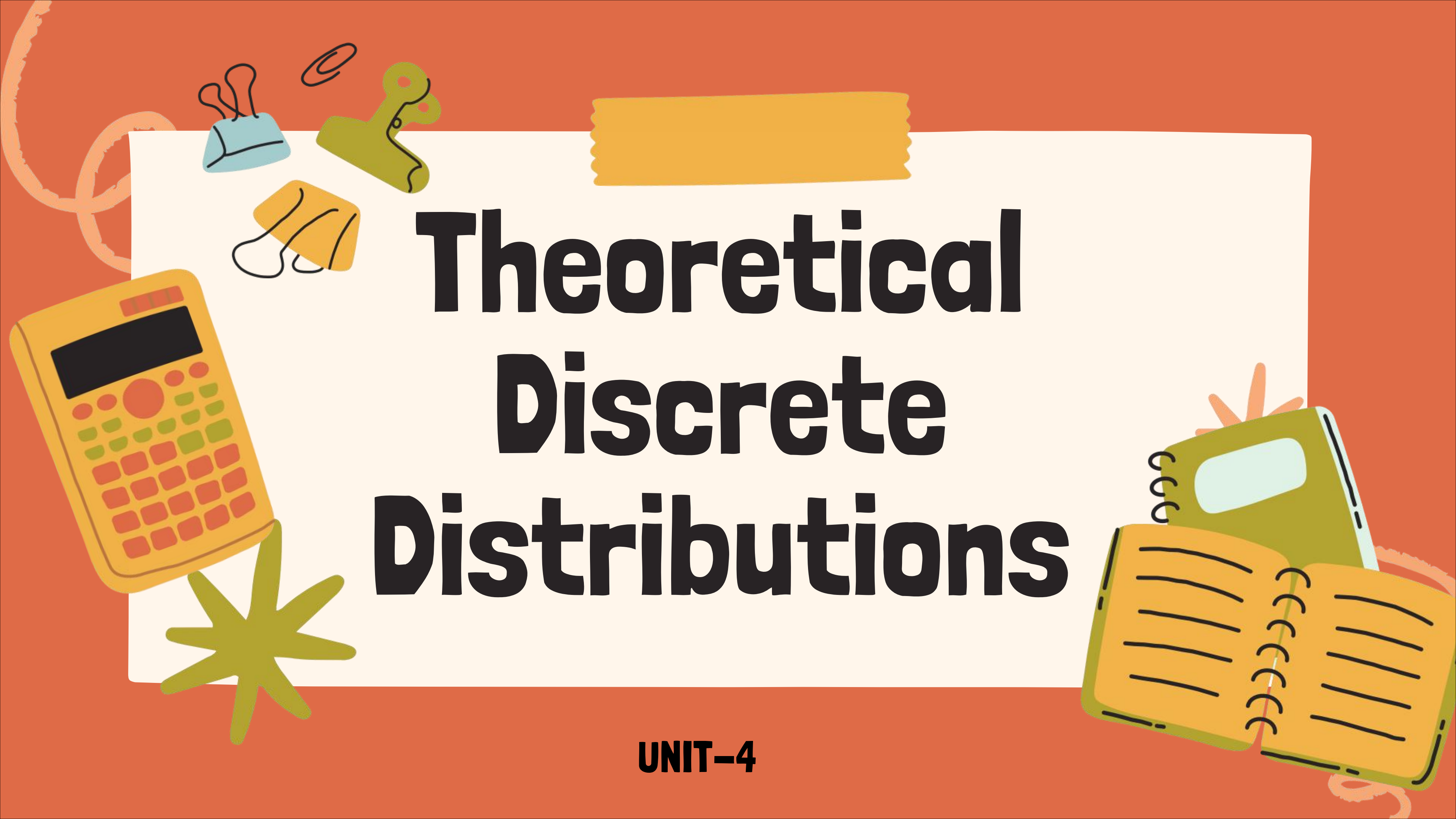
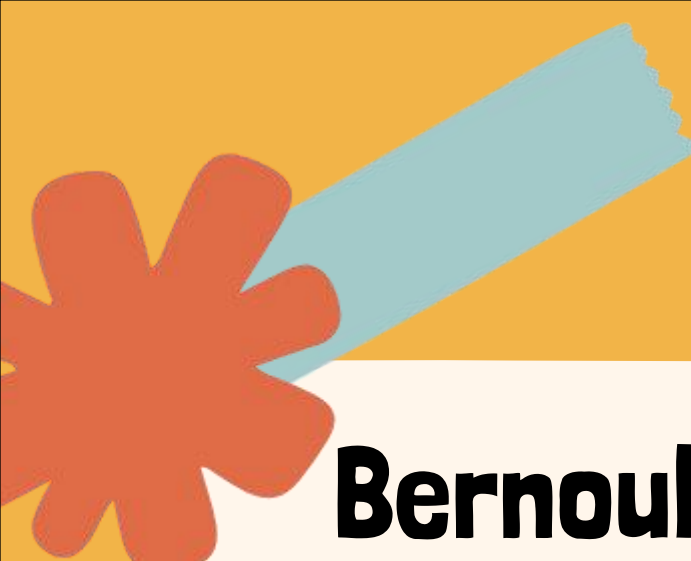




Theoretical Discrete Distributions

UNIT-4



Bernoulli Distribution

Any random experiment whose outcomes can be classified only into 2 types as (success **S** and Failure **F**) is called Bernoulli trial

CONDITIONS:

- The number of trials should be finite.
- Each trial should be independent.
- Each trial should have only two possible outcomes - success and failure.
- The probability of each outcome should be the same in every trial.



Defination:

A discrete random variable X is said to follow Bernoulli distribution if its probability mass function is given by

$$P(X=x) = p^x(1-p)^{(1-x)} ; x=0,1$$

0

o.w

Where

- p is the probability of success and $(1-p)=q$ is the probability of failure
- p is the parameter of the distribution and $0 < p < 1$
- Bernoulli distribution is written as $X \sim \text{Bernoulli}(1,p)$



Defination:

- A newborn child is either male or female. (Here the probability of a child being a male is roughly 0.5.)
- You either pass or fail an exam.
- A tennis player either wins or loses a match.



Mean and Variance of Bernoulli Distribution

$$\begin{aligned}E(X) &= \sum_{x=0}^1 xP(X=x) \\&= \sum_{x=0}^1 xp^x(1-p)^{1-x} \\&= 0p^0(1-p)^{1-0} + 1p^1(1-p)^{1-1} \\ \therefore \mathbf{E(X) = p}\end{aligned}$$

$$\begin{aligned}E(X^2) &= \sum_{x=0}^1 x^2P(X=x) \\&= \sum_{x=0}^1 x^2p^x(1-p)^{1-x} \\&= 0p^0(1-p)^{1-0} + 1p^1(1-p)^{1-1} \\&= p\end{aligned}$$

Now,

$$\begin{aligned}\mathbf{Var(X)} &= \mathbf{E(X^2) - [E(X)]^2} \\&= p - p^2 \\&= p(1-p)\end{aligned}$$

$$\mathbf{Var(X) = pq}$$



Binomial Distribution

Consider a set of n independent Bernoullian trials (n being finite) in which the probability 'p' of success in any trial is constant for each trial, and then $q=1-p$ is the probability of failure in any trial.

The probability of x successes and consequently $(n - x)$ failures in n independent trials, in a specified order (say) **SSFSFFFS...FSF** (where **S** represents success and **F** represents failure) is given by the compound probability theorem by the expression:

$$\begin{aligned} P(\text{SSFSFFFS...FSF}) &= P(S)P(S)P(F)P(S)P(F)P(F)P(F)P(S) \times \dots \times P(F)P(S)P(F) \\ &= p.p.q.p.q.q.q.p \dots q.p.q \\ &\quad \{x \text{ factors}\} \quad \{(n-x) \text{ factors}\} \end{aligned}$$





But x successes in n trials can occur in $\binom{n}{x}$ ways and the probability for each of these ways is same, viz, $p^x q^{n-x}$

Hence the probability of x successes in n trials in any order is given by the addition theorem of probability by the expression $\binom{n}{x} p^x q^{n-x}$

The probability distribution of the number of successes, so obtained is called the Binomial probability distribution.





Defination:

A random variable X is said to follow binomial distribution if it assumes only non-negative values and its probability mass function is given by

$$P(X = x) = p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x} & ; x = 0, 1, 2, \dots, n \quad ; q = 1 - p \\ 0 & , otherwise \end{cases}$$

Notation:

It is represented as $X \sim \text{Bin}(n, p)$ i.e. random variable X follows Binomial distribution with parameters n and p





Conditions For Binomial Distribution

1. Each trial results in two exhaustive and mutually disjoint outcomes, termed as success and failure.
2. The number of trials ' n ' is finite.
3. The trials are independent of each other.
4. The probability of success ' p ' is constant for each trial.





A fair coin is tossed five times. Find the probability of getting

- a) 3 heads and 2 tails
- b) Exactly 5 heads
- c) At least 3 heads

A coin is tossed 5 times

$$\therefore n = 5$$

Let getting head is called as success

$$\therefore p = \text{probability of success} = \frac{1}{2}$$

$$\begin{aligned} \& q = \text{probability of failure in each trial} = 1 - p \\ &= 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

Let X denote number of heads

$$\therefore \text{Possible values of } X \text{ are } x = 1, 2, 3, 4, 5$$

$$\therefore X \sim B(n, p)$$

$$\therefore X \sim B\left(5, \frac{1}{2}\right)$$

$$\therefore P(X = x) = {}^n C_x p^x q^{n-x}, x = 1, 2, 3, 4, 5$$





i. $p(\text{getting 3 heads and 2 tails}) = p(\text{getting 3 success})$

$$= P(X = 3)$$

$$= {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2$$

$$\therefore p(\text{getting 3 heads and 2 tails}) = 0.3125$$

ii. $p(\text{exactly 5 heads}) = p(X=5)$

$${}^5C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^0 = \frac{1}{32} = 0.03125$$

iii. $p(\text{at least 3 heads}) = p(X \geq 3)$

$$= p(X = 3) + p(X = 4) + p(X = 5)$$

$$= {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 + {}^5C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^1 + {}^5C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^0$$

$$= \frac{5}{16} + \frac{5}{32} + \frac{1}{32} = \frac{1}{2}$$

$$p(\text{at least 3 heads}) = 0.5$$





Q.2 A and B play a game in which their chances of winning are in the ratio 3:2. Find A's chance of winning at least 3 games out of the 5 games played

Let p be the probability that A wins the games. Then we are given $p = \frac{3}{5}$

$$\therefore q = 1 - p = 1 - \frac{3}{5} = \frac{2}{5}$$

$$n=5$$

$$\therefore X \sim B(n, p) = B\left(5, \frac{3}{5}\right)$$

$$P(X = x) = {}^nC_x p^x q^{n-x}, x = 1, 2, 3, 4, 5$$

$\therefore$ The probability that A wins at least 3 games is given by ,

$$p(X \geq 3) = p(X = 3) + p(X = 4) + p(X = 5)$$

$$= {}^5C_3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)^2 + {}^5C_4 \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right)^1 + {}^5C_5 \left(\frac{3}{5}\right)^5 \left(\frac{2}{5}\right)^0$$

$$= \frac{2133}{3125}$$

$$p(X \geq 3) = 0.68256$$





Practice Questions

1. A lot of 100 pens contains 10 defective pens. 5 pens are selected at random from the lot and sent to the retail store, what is the probability that the store will receive at least 1 defective pen?
2. Ten coins are thrown simultaneously. Find the probability of getting at-least seven heads
3. The mean and Variance of Binomial Distribution is 4 and $\frac{4}{3}$ respectively. Find $P(X \geq 1)$
4. If X has Binomial Distribution with $n=20$ $p=\frac{1}{2}$. Find the mean and variance of X



Homework Questions

Q.1 A maker of specialized instruments receives shipments of 24 circuit boards. Suppose one shipment contains 4 that are defective. An engineer selects a random sample of size 4. What are the probabilities that the sample will contain

- (a) 0 defective circuit boards?
- (b) 1 defective circuit board ?
- (c) 2 or more defective circuit boards?

Q.2 If 6 of 18 new buildings in a city violate the building code, what is the probability that a building inspector,

who randomly selects 4 of the new buildings for inspection, will catch

- (a) none of the buildings that violate the building code?
- (b) 1 of the new buildings that violate the building code?
- (c) 2 of the new buildings that violate the building code?
- (d) at least 3 of the new buildings that violate the building code?

Q.4 A hotel has four elevators. One of them is a freight elevator. When pressing the button, one of the elevators randomly services your floor. If you use the elevators five times, what is the probability that you use the freight elevator exactly two times?

Q.5 A test consists of eleven true/false questions. A student who forgot to study guesses randomly on every question. What is the probability that the student answers at least ten questions correctly?



For a binomial variate X , means and variance are 2 and 1 respectively.
Find $P(X=3)$

Let $X \sim B(n, p)$

Mean = $np = 2$

Variance = $npq = 1$

$\therefore 2q = 1$ ----- since $np = 2$

$\therefore q = \frac{1}{2}$

$\therefore p = 1 - q = 1 - \frac{1}{2} = \frac{1}{2}$

Since $np = 2$

$\therefore n * \frac{1}{2} = 2$

$\therefore n = 4$

$\therefore X \sim B(n, p) = X \sim B\left(4, \frac{1}{2}\right)$

$P(X = x) = {}^nC_x p^x q^{n-x}$

$P(X = 3) = {}^4C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2$

$P(X = 3) = 0.25$





Recurrence Relation for Probabilities of Binomial Distribution (Fitting of Binomial Distribution)

$$\begin{aligned}\frac{P(X=x+1)}{P(X=x)} &= \frac{\binom{n}{x+1} p^{x+1} q^{n-(x+1)}}{\binom{n}{x} p^x q^{n-x}} \\ &= \frac{\frac{n!}{(x+1)!(n-x-1)!} \times p^x \times p \times q^{n-x} \times q^{-1}}{\frac{n!}{x!(n-x)!} \times p^x \times q^{n-x}} \\ &= \frac{(n-x)p}{(x+1)q}\end{aligned}$$

$$\therefore P(X = x + 1) = \frac{(n-x)p}{(x+1)q} P(X = x)$$

This is recurrence formula.

If $P(0)$ is given we can find the remaining probabilities $P(1), P(2), \dots$

When $P(0)$ is not given we can use $P(0) = q^n$

q^n can be computed from given data by equating mean $\bar{x}$ to np





Fit a binomial distribution for the following data and hence find the expected frequencies

X	0	1	2	3	4
f	5	29	36	25	5

$$\text{Mean} = \bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

$$\bar{x} = \frac{5(0) + 29(1) + 36(2) + 25(3) + 5(4)}{5 + 29 + 36 + 25 + 5}$$

$$\bar{x} = \frac{196}{100} = 1.96$$

$$\text{Here } n=5 ; N = \sum f_i = 100$$

$$\text{For binomial distribution mean} = \bar{x} = np$$

$$\therefore 1.96 = np \text{ and } n = 5$$

$$\therefore \frac{1.96}{n} = p = \frac{1.96}{5}$$

$$\therefore p = 0.39$$





$$\therefore q = 1 - p = 1 - 0.39$$

$$\therefore q = \mathbf{0.608}$$

$$\therefore X \sim B(n, p) = X \sim B(5, 0.49)$$

$$\begin{aligned} P(X = x) &= {}^nC_x p^x q^{n-x}, \quad x = 0, 1, 2, 3, 4 \\ &= {}^5C_x (0.49)^x (0.51)^{5-x}, \quad x = 0, 1, 2, 3, 4 \end{aligned}$$

For $x = 0$

$$\begin{aligned} P(X = 0) &= {}^5C_0 (0.39)^0 (0.608)^{5-0} = {}^5C_0 (0.39)^0 (0.608)^5 \\ &= 0.08308 \end{aligned}$$

For $x = 1$

$$\begin{aligned} P(X = 1) &= {}^5C_1 (0.39)^1 (0.608)^{5-1} = {}^5C_1 (0.39)^1 (0.608)^4 \\ &= 0.2678 \end{aligned}$$

For $x = 2$

$$\begin{aligned} P(X = 2) &= {}^5C_2 (0.39)^2 (0.608)^{5-2} = {}^5C_2 (0.49)^2 (0.608)^3 \\ &= 0.345 \end{aligned}$$

For $x = 3$

$$\begin{aligned} P(X = 3) &= {}^5C_3 (0.39)^3 (0.608)^{5-3} = {}^5C_1 (0.49)^3 (0.608)^2 \\ &= 0.2222 \end{aligned}$$





For $x = 4$

$$P(X = 4) = {}^5C_4(0.39)^4(0.608)^{5-4} = {}^5C_4(0.39)^4(0.608)^1 \\ = 0.072$$

Now Expected frequency = $f = N * P(X = x) = 100 * P(X = x)$

$$\text{For } x=0, f = 100 * P(X = 0) = 100 * 0.0830 \approx 8$$

$$\text{For } x=1, f = 100 * P(X = 1) = 100 * 0.2678 \approx 27$$

$$\text{For } x=2, f = 100 * P(X = 2) = 100 * 0.345 \approx 35$$

$$\text{For } x=3, f = 100 * P(X = 3) = 100 * 0.2222 \approx 22$$

$$\text{For } x=4, f = 100 * P(X = 4) = 100 * 0.072 \approx 7$$

$\therefore$ Fitting of Binomial distribution is ,

x	0	1	2	3	4
f	8	27	35	22	7





Fit a binomial distribution to the following data

X	0	1	2	3	4	5
f	36	39	18	4	2	1



Poisson Distribution

A random variable X is said to follow Poisson distribution if it assumes only non-negative values and its Probability Mass Function(PMF) is given as

$$p(x, \lambda) = P(X = x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}; & x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

Here λ is known as the parameter of the distribution. It is denoted by $X \sim P(\lambda)$ i.e., X is a Poisson variate with parameter .

Conditions for Poisson Distribution

1. No of trials are infinite
2. Events are independent of each other
3. It is used for modelling rare events
4. The Poisson distribution assumes that the rate at which events occur is constant over time or space.
5. The Poisson distribution is only suitable for modeling count data

Note:

Binomial Distribution can be approximated to Poisson if Sample size is large i.e $n \rightarrow \infty$ and p is infinitesimally small $p \rightarrow 0$

Mean and Variance for Poisson Distribution

$$\text{Mean} = E[X] = \sum_{x=0}^{\infty} x p(x, \lambda) = \lambda$$

$$\text{Variance} = V[X] = \lambda$$

Example 1

A car hire firm has two cars which it hire on the day to day basis. The number of demand for a car on each day is distributed as Poisson variate with mean is 1.5. Calculate the proportion of days on which;

- i. Neither car is used
- ii. Some demand is refused

$$\text{Mean} = \lambda = 1.5$$
$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, 3 \dots$$

$$P[X = x] = \frac{e^{-1.5} 1.5^x}{x!}; x = 0, 1, 2, 3 \dots$$

$$\begin{aligned} i. \quad \therefore P(\text{neither car used}) &= P(X = 0) \\ &= \frac{e^{-1.5} 1.5^0}{0!} \end{aligned}$$

$$= 0.223130$$

$$\begin{aligned} ii. \quad \therefore P(\text{some demand is refused}) &= P(X > 2) \\ &= 1 - P(X \leq 2) \\ &= 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\ &= 1 - \left[\frac{e^{-1.5} 1.5^0}{0!} + \frac{e^{-1.5} 1.5^1}{1!} + \frac{e^{-1.5} 1.5^2}{2!} \right] \\ &= \mathbf{0.191153} \end{aligned}$$

Practice Questions

1. It is observed that average number of phone calls per minute coming into a switchboard of a company is 3. Find the probability that during a particular minute there will be

- i) No phone calls
- ii) Exact 2 phone calls
- iii) Atleast 3 phone calls.

2. It is observed that 2% of bulbs made by a factory are defective. Find probability that in a sample of 200 bulbs i) less than 2 bulbs ii) more than 3 bulbs are defectives.

3. A manufacturer of pins knows that 5% of his product is defective. If he sells pins in boxes of 100 and guarantees that not more than 10 pins will be defective, what is the approximate probability that a box will fail to meet the guaranteed quality?

4. Suppose that the number of telephone calls coming into a telephone exchange between 10 A.M. and 11 A.M. is x_1 and the number of calls arriving between 11 A.M. and 12 noon is x_2 where x_1 and x_2 are random variables of Poisson distribution with parameter 2 and 6 respectively. If x_1 and x_2 are independent what is the probability that more than 5 calls come in between 10 A.M. and 12 noon?

Recurrence Formula for the probabilities of Poisson Distribution(Fitting o Poisson Distribution):

For a Poisson distribution with parameter λ we have,

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots$$

We can write,

$$P[X = x + 1] = \frac{e^{-\lambda} \lambda^{(x+1)}}{(x+1)!}, x = 0, 1, 2, \dots$$

Therefore,

$$\frac{P[X = x + 1]}{P[X = x]} = \frac{\frac{e^{-\lambda} \lambda^{(x+1)}}{(x+1)!}}{\frac{e^{-\lambda} \lambda^x}{x!}}, x = 0, 1, 2, \dots$$

$$\Rightarrow P[X = x + 1] = \frac{\lambda}{x+1} P[X = x]$$

Example 1:

Fit a poisson distribution for the following data .

x	0	1	2	3	4
f	32	45	15	7	1

Solution:

$$\text{Mean}=\lambda = \frac{\sum f_i x_i}{\sum f_i}$$

$$\lambda = \frac{32(0) + 45(1) + 15(2) + 7(3) + 1(4)}{32 + 45 + 15 + 7 + 1}$$

$$\lambda = \frac{100}{100} = 1$$

Poisson distribution is ,

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots$$

Expected frequency = $f = N.P$,

$$\text{Here, } N = \sum f_i = 100, f = 100 * P(X = x)$$

$$\therefore f = \frac{100 * e^{-1}(1)^x}{x!} \text{ ----- } x = 0,1,2,3,4$$

$$\text{For } x = 0; f = \frac{100 * e^{-1}(1)^0}{0!} = 36.79 \approx 37$$

$$\text{For } x = 1; f = \frac{100 * e^{-1}(1)^1}{1!} = 36.79 \approx 37$$

$$\text{For } x = 2; f = \frac{100 * e^{-1}(1)^2}{2!} = 18.40 \approx 18$$

$$\text{For } x = 3; f = \frac{100 * e^{-1}(1)^3}{3!} = 6.13 \approx 6$$

$$\text{For } x = 4; f = \frac{100 * e^{-1}(1)^4}{4!} = 1.53 \approx 2$$

Fitting of Poisson distribution is ,

x	0	1	2	3	4
f	37	37	18	6	2

Questions for Practice:

1. After correcting 50 pages of the proof of a book, the proof reader finds that there are on an average 2 errors per 5 pages. How many pages would one expect to find with 0, 1, 2, 3 and 4 errors in 1000 pages of the book?

2. Fit Poisson distribution for the following data. Find probabilities using recurrence relation.

x	1	2	3	4	5
f	32	45	15	7	1



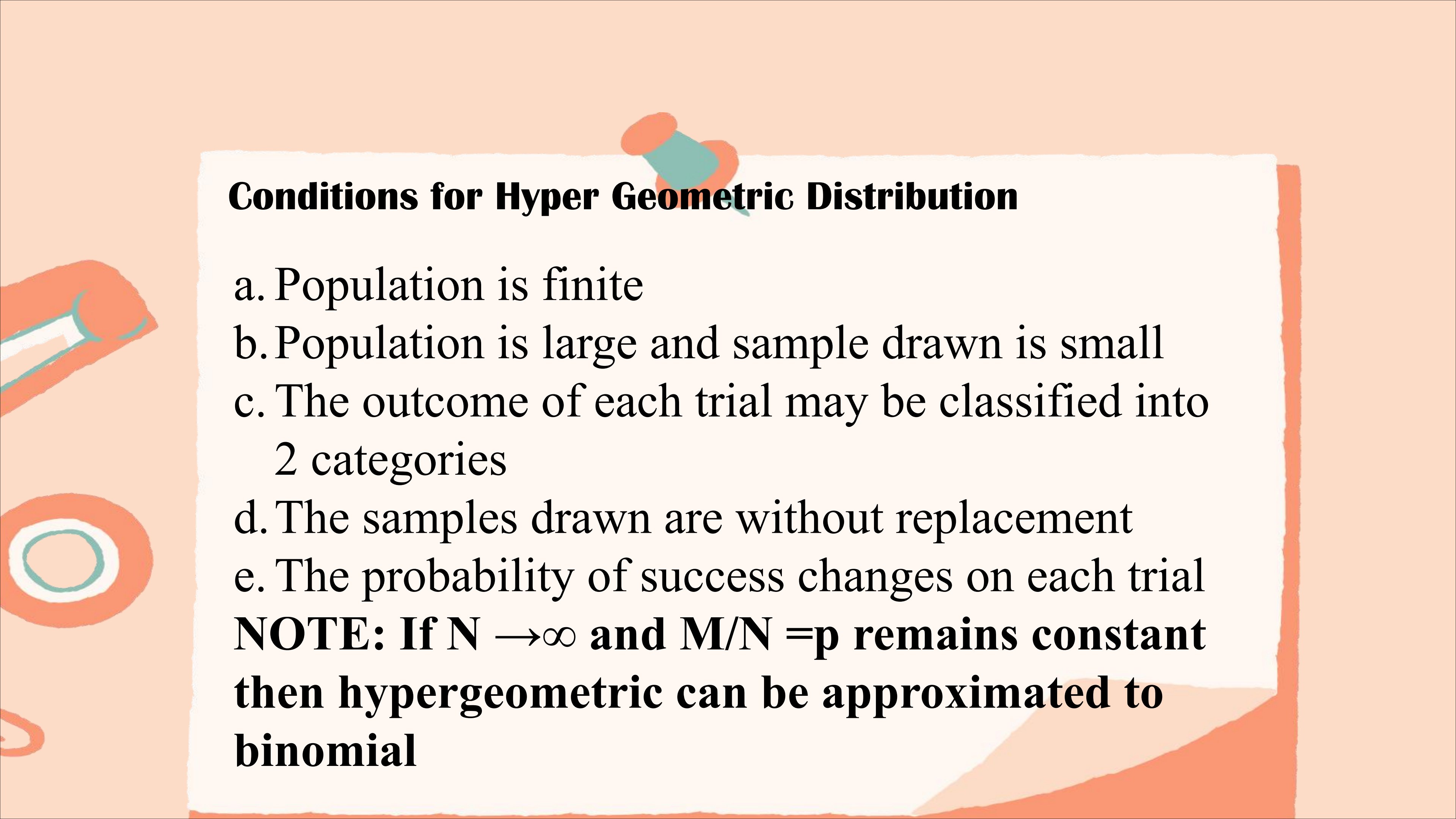
Hyper Geometric Distribution

A discrete random variable X is said to follow hypergeometric distribution if it assumes only non-negative values and its probability mass function is given by

The distribution given by

$$P(X = r) = \frac{{}^M C_r \times {}^{N-M} C_{n-r}}{{}^N C_n}$$

which describes the probability of obtaining a sample of size n containing r defective items from a population of size N known to contain M defective items is known as the **hypergeometric distribution**.



Conditions for Hyper Geometric Distribution

- a. Population is finite
- b. Population is large and sample drawn is small
- c. The outcome of each trial may be classified into 2 categories
- d. The samples drawn are without replacement
- e. The probability of success changes on each trial

NOTE: If $N \rightarrow \infty$ and $M/N = p$ remains constant then hypergeometric can be approximated to binomial

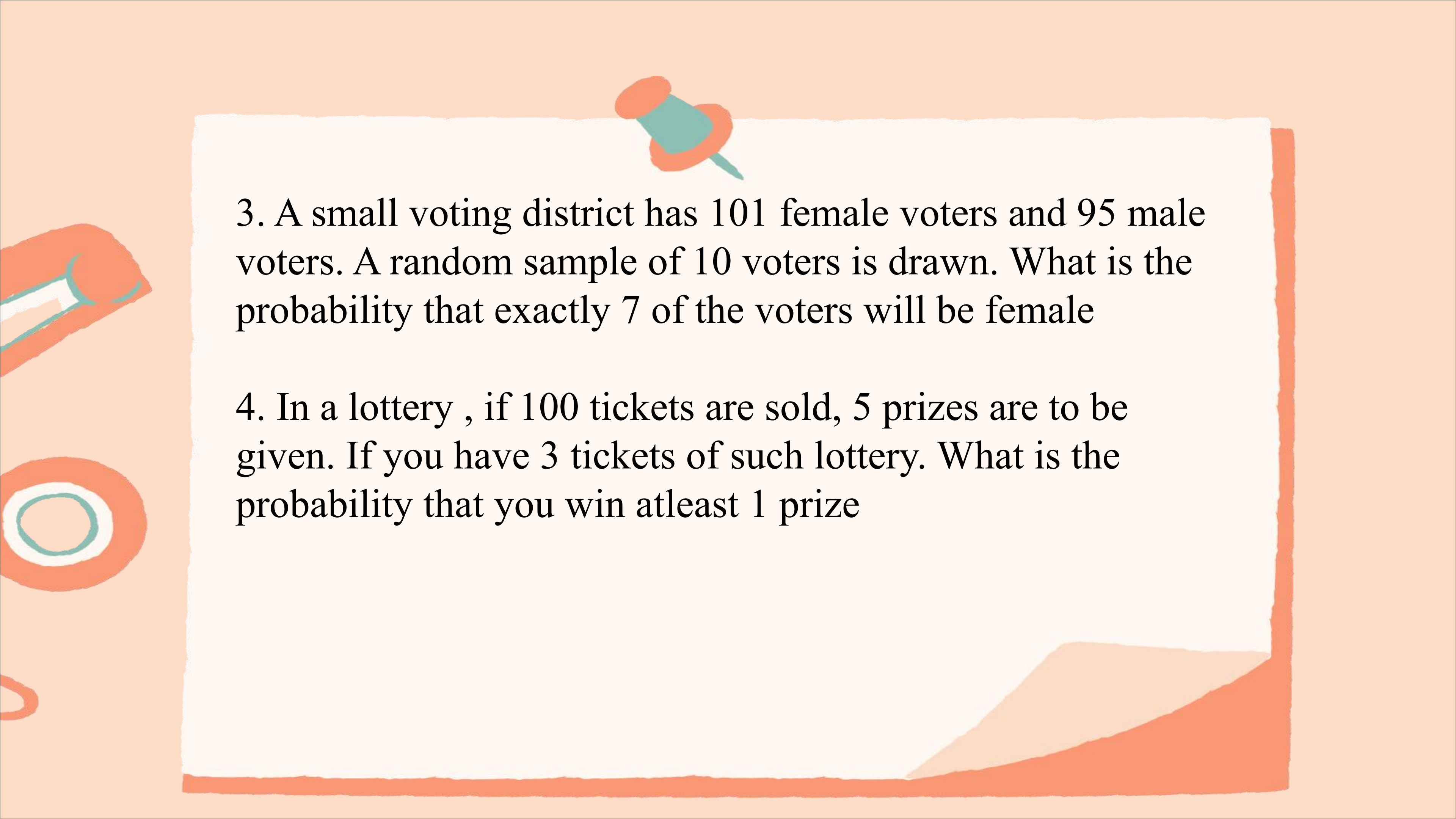


1. Five police officers and 20 firefighters attend a small conference. All 25 names are placed in a hat and 5 names are drawn without replacement. What is the probability that 3 police officers and 2 firefighters are chosen



2. Boxes contain 2000 items of which 10% are defective Find the probability that 2 defectives will be obtained in a sample of size 10





3. A small voting district has 101 female voters and 95 male voters. A random sample of 10 voters is drawn. What is the probability that exactly 7 of the voters will be female

4. In a lottery , if 100 tickets are sold, 5 prizes are to be given. If you have 3 tickets of such lottery. What is the probability that you win atleast 1 prize



Geometric Distributions:

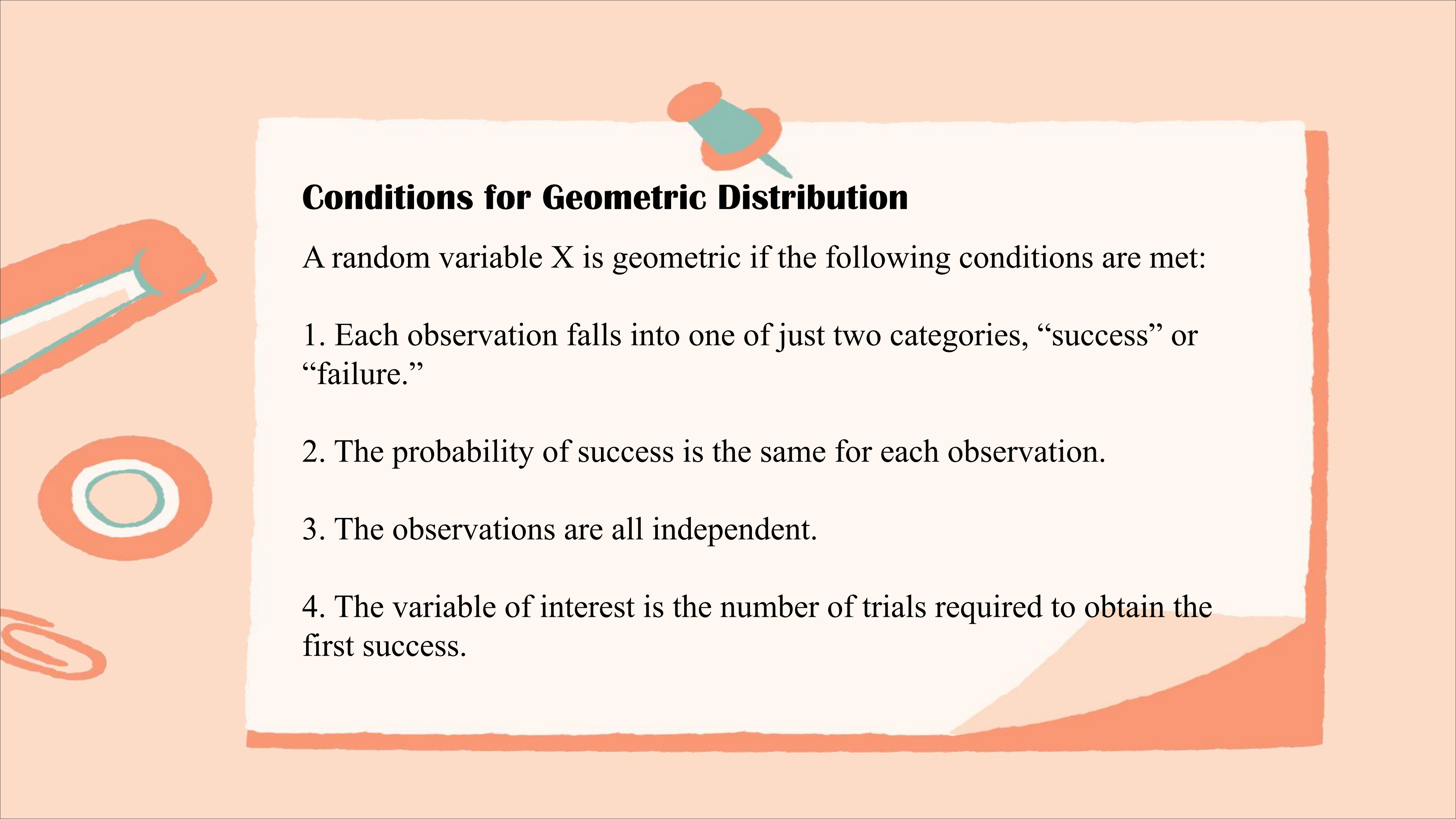


The Geometric Distribution is a discrete probability distribution. It is the distribution of the number of trials required to get the first success in repeated Bernoulli trials

The geometric distribution defines the probability that first success occurs after x number of trials. p is the probability of success or failure

$$P(X = x) = (1 - p)^{x-1} p ; x=1,2,3,\dots$$

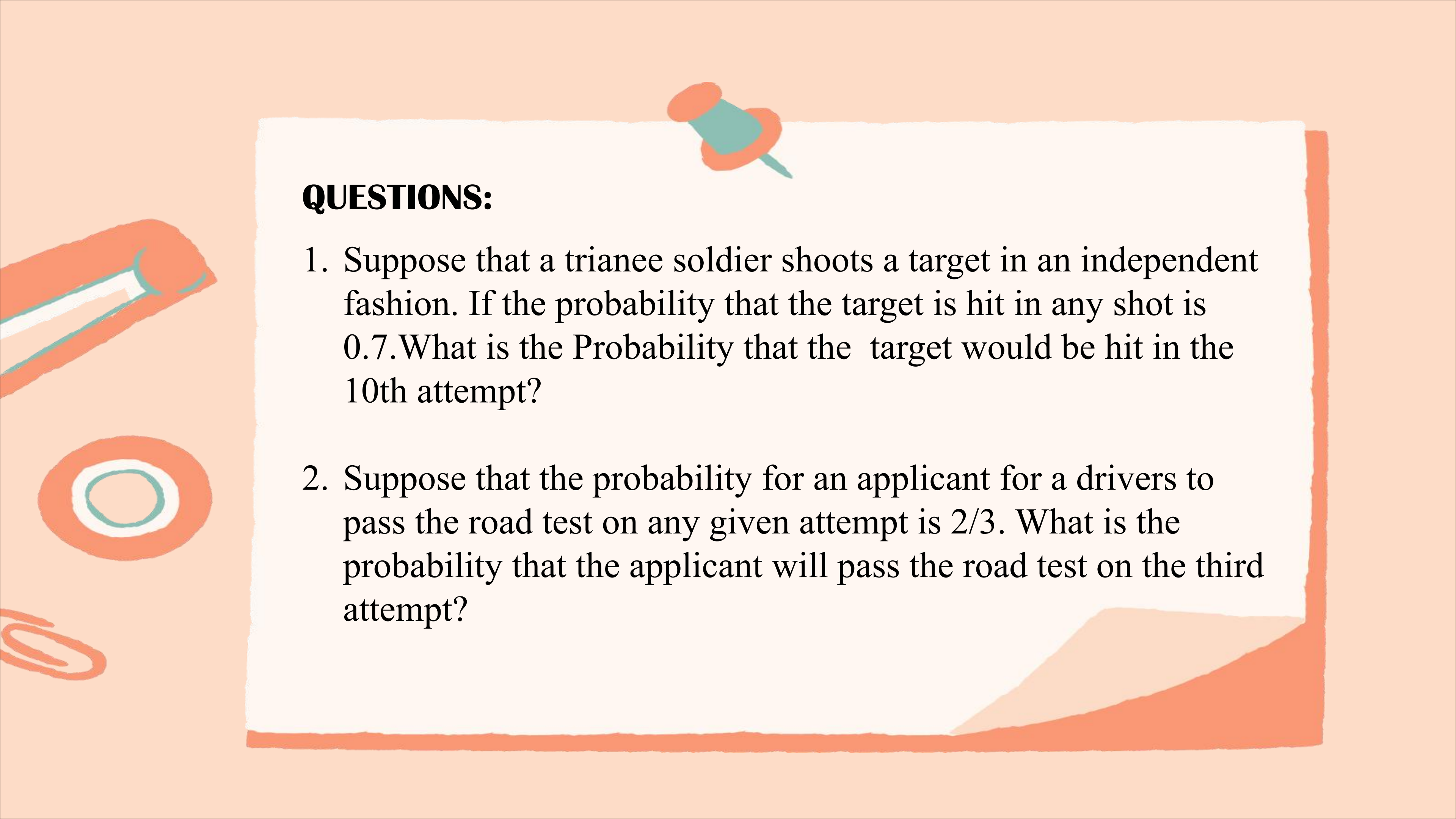
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Conditions for Geometric Distribution

A random variable X is geometric if the following conditions are met:

1. Each observation falls into one of just two categories, “success” or “failure.”
2. The probability of success is the same for each observation.
3. The observations are all independent.
4. The variable of interest is the number of trials required to obtain the first success.



QUESTIONS:

1. Suppose that a trainee soldier shoots a target in an independent fashion. If the probability that the target is hit in any shot is 0.7. What is the Probability that the target would be hit in the 10th attempt?
2. Suppose that the probability for an applicant for a drivers to pass the road test on any given attempt is $\frac{2}{3}$. What is the probability that the applicant will pass the road test on the third attempt?



Multinomial Distribution

The multinomial distribution describes the probability of obtaining a specific number of counts for k different outcomes, when each outcome has a fixed probability of occurring

Probability = $n! * \frac{(p_1^{x_1} * p_2^{x_2} * ... * p_k^{x_k})}{(x_1! * x_2! * ... * x_k!).}$ **where:**

n : total number of events

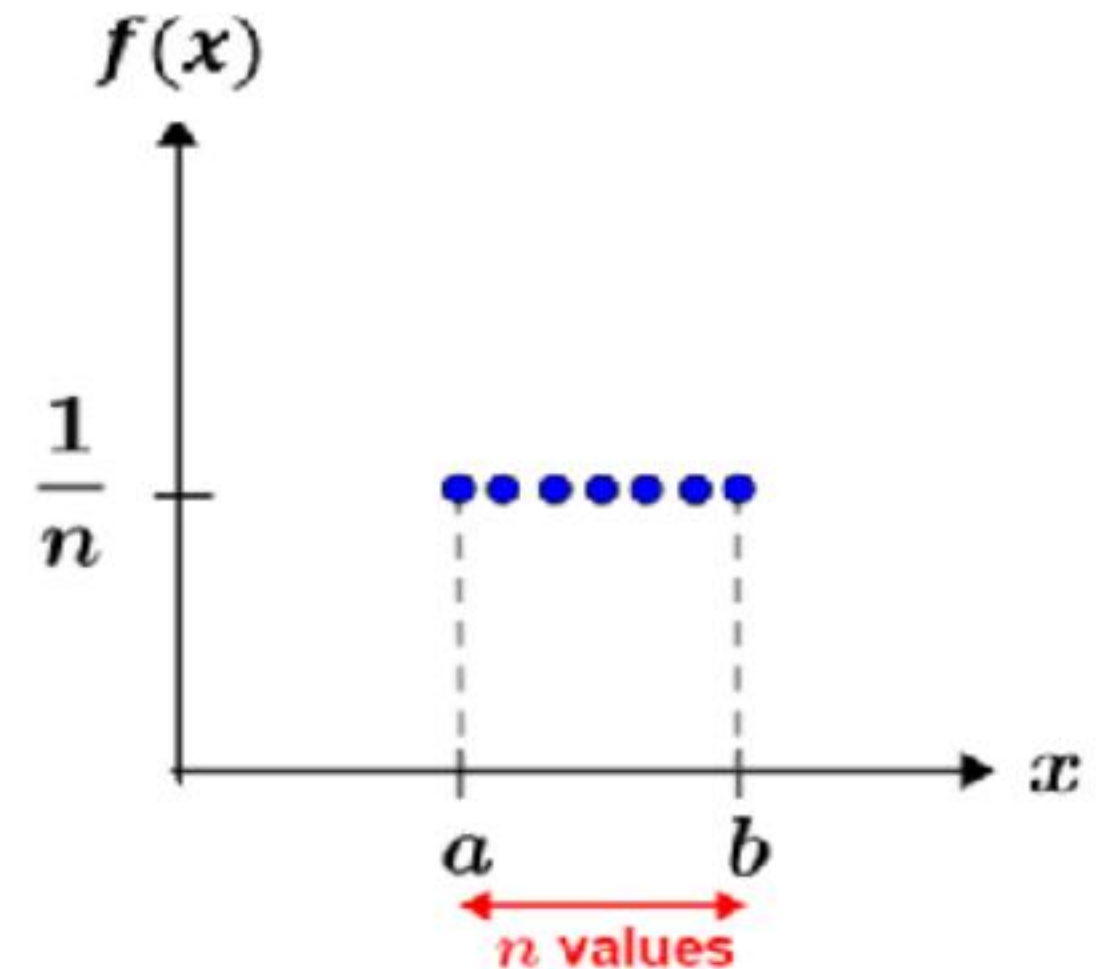
x_1 : number of times outcome 1 occurs

p_1 : probability that outcome 1 occurs in a given trial

Discrete Uniform Distributions

A discrete random variable X can assume n distinct values and every one of the n distinct values has equal probability $1/n$ we are dealing with a discrete uniform distribution

$$f(x) = \frac{1}{n}, \quad x = 1, 2, 3, \dots, n$$





Examples:



A fair coin is tossed once. Then the probability of Getting head is ?



A single dice is rolled. What is the probability of number on the uppermost face.

